Pavana Naga Lakshmi Sankari Penumaka

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Data Analyst with around 2 years of experience with a strong foundation in data analysis, developing data visualizations, and implementing automation solutions to drive efficiency and productivity, resulting in improved performance and strategic decision making.

Education

University of North Carolina at Charlotte May 2024
Master of Science in Informatics and Analytics Charlotte, NC

Skills

- **Programming**: Python, R, SQL, SAS
- **Databases**: MySQL, PostgreSQL, Snowflake, MS SQL Server
- **Analytical Tools**: MS Excel, Power BI, Tableau, Alteryx
- **Libraries :** NumPy, Pandas, Matplotlib, Flask, Scikit-learn
- **Tools & Technologies:** AWS, EC2, Git, GitHub, Asana, Jira
- **Methodologies**: Agile, Waterfall, Kanban

Experience

Data Science Intern January 2024 – Present
Camino Research Institute

- Utilized **NumPy** for efficient numerical operations and **pandas** for **data manipulation** to analyze and cleaned over **100+ data points** for the Western North Carolina Latino Community Strengths and Needs Assessment, ensuring accuracy and reliability.
- Employed **scikit-learn** for regression analysis on a dataset with over 30 variables, adjusting for specific demographic factors.
- Guided the development of a statewide Latino Community Strengths and Needs Assessment dashboard through **Tableau**, incorporating 20+ key data visualizations for effective communication.

Graduate Teaching Assistant - Data Science and Analytics August 2023 - May 2024
University of North Carolina at Charlotte

- Showcased 10+ research projects and Utilized **SQL** for data handling to direct 10+ research presentations at departmental seminars contributing to the advancement of knowledge in Informatics.
- Managed 40+ hours of hands-on lab sessions weekly and leveraged **Power BI** for data visualization, leading to a **30% improvement** in student performance and project outcomes.
- Provided personalized leadership and guidance to 30+ students, fostering a culture of academic excellence and professional development.

Research Data Analyst January 2021 – July 2022
Sujay Biotech Pvt Ltd

- Executed clinical trial data from 20+ studies using **SAS and R**, ensuring compliance with regulatory standards and leading detailed project documentation, improved data accuracy and project efficiency.
- Structured and analyzed patient outcome data in **Microsoft Excel**, uncovering critical insights that reduced time-to-market for new therapies by 15% and improved treatment efficacy by 10%.
- Conducted **in-depth trend analysis** on healthcare datasets using SAS and R; identified key market opportunities, informing product development strategies, and increasing potential market share by 20%.

Projects

Business Analysis Insights Dashboard [Power BI | SQL | Excel | Dax Studio]

- Designed a **Power BI dashboard** gathering sales data from 2 different sources (Excel/CSV files and a SQL database).
- Optimized with **DAX Studio**, resulting in a 5% performance boost and projected 10% revenue growth and 20% data-related expense reduction.

Unveiling Insights into Bank Lending with Data Analytics [MS SQL Server | Tableau]

- Utilized **MS SQL** to clean and analyze over 10,000 loan records, ensuring **data accuracy** and reliability. Developed 15+ dynamic visualizations in **Tableau**, providing critical insights into lending trends and portfolio health.
- Created a comprehensive dashboard that empowers 50+ stakeholders to make informed, data-driven decisions, driving financial stability and growth.

Hospitality Data Analysis - [Python]

- Utilized Python (**Pandas, NumPy, Matplotlib**) to analyze data, extracting occupancy, revenue, and booking preferences trends. Implemented robust data cleaning and transformation to maintain data integrity, **reducing error rates by 20%.**

- Produced actionable reports assisting hotel management in strategic planning and optimization, projecting a 15% increase in revenue and a 10% improvement in occupancy rates.

Building Virtual Assistant and Email Automation [Python | SQL]

- Developed a **Python virtual Assistant** with 90% accuracy in speech recognition, integrating gTTS, speech recognition, and playsound libraries. Ensured project status updates were regularly communicated to stakeholders.
- Automated email sending, testing, web search, and location mapping using **Google APIs**, reducing manual tasks by 20% through enhanced assistant functionalities.

Coca-Cola Retailer Dashboard [MS Excel | Pivot Tables | VLOOKUP]

- Cleaned and analyzed Coca-Cola retailer dataset of 10,000+ entries using Excel, removing duplicates, filling missing values, and correcting data formats.
- Created an interactive dashboard with pivot tables to visualize key metrics, performing time series analysis and comparative analysis across 5 regions and 10 product categories.

Certifications

- Data Analytics Essentials certification by **Cisco**[Link]
- Alteryx Designer Core certification by **Alteryx[Link]**
- Associate Data Analyst in SQL Certification by **Data camp[Link]**
- **Accenture** - Data Analytics and Visualization Job Simulation by **Forage**[Link]
- Social & Behavioral Research (IRB) certification by **CITI** Program[Link]

Peer Reviewed Publications – [Publication]

A Data-Driven Approach to Defining Risk-Adjusted Coding Specificity Metrics for a Large U.S. Dementia Patient Cohort

- Deployed **R and Posit Cloud** to analyze 487,775 hospitalization records, achieving a 72.7% accuracy in predicting principal dementia diagnoses through **logistic regression models.**
- Implemented **novel risk-adjusted metrics** and **geospatial visualizations**, resulting in a 15% improvement in coding accuracy across facilities and enhancing patient care. Accompanied forecasting to predict future trends in coding accuracy.